Experimental Benchmarking Plan & Evaluation Protocol

Paper Title: *A Neuro-Symbolic Agentic Framework for Autonomous Structural Modeling, Code-Compliant Design, and Detailing from Multimodal Inputs*
Target Venue: *Automation in Construction / Advanced Engineering Informatics*
Companion Blueprint: `research_paper/paper_idea.md`

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1. Experimental Objectives

The experimental campaign is designed to rigorously test Hypotheses H1–H3 across five core dimensions:

1. **Numerical Parity & Solver Trust**: Verifying that the deterministic symbolic solvers (`fem2d`, `fem3d`, `structdyn`, `aci318`) match commercial CSI SAP2000/ETABS and analytical benchmarks within $< 0.5\%$.
2. **Determinism & Hallucination Elimination (H1)**: Confirming a 0.0% Calculation Hallucination Rate (CHR) compared to direct zero-shot LLM prompting and raw script generation.
3. **Self-Healing Convergence (H2)**: Measuring the ability of the closed-loop reflection mechanism to autonomously resize initially failing or overstressed structural members ($\mathrm{DCR}_0 > 1.0 \rightarrow \mathrm{DCR}_{\text{final}} \leq 1.0$) under static, Response Spectrum, and earthquake time-history loading.
4. **Multi-Model Generalization (H3)**: Evaluating robustness across frontier closed-source APIs (`gemini-3.8-flash`) and local/cloud open-weight models via Ollama (`qwen3.5:9b`, `gpt-oss:120b-cloud`, `nemotron-3-ultra:cloud`).
5. **Design Quality & Material Economy (H2)**: Comparing the agent's converged concrete volumes ($V_{\text{concrete}}^{\text{agent}}$) and turnaround times against verified ACI 318-compliant human baseline designs ($V_{\text{concrete}}^{\text{baseline}}$).

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2. Standardized 24-Case Benchmark Suite (`benchmark_cases.json`)

The benchmark suite consists of **24 structural engineering problems** divided into three progressive groups (18 cases requiring full ACI 318 code compliance verification and 6 cases evaluating pure modal dynamics, 3D kinematics, vision compilation, or CAE round-trip interoperability):

Group A: 2D Plane Frames (8 Cases) — Gravity, Modal Dynamics & Response Spectrum

- **A01 (`A01_Portal_Frame_Office`)**: Single-Bay RC Portal Frame ($L = 6.0\ \mathrm{m}, H = 3.5\ \mathrm{m}$) under ASCE 7 office gravity loads ($w_{\text{trib}} = 4.0\ \mathrm{m}$), fixed base, ACI 318 design, and production DXF drawing generation saved to `research_paper/drawings/A01_portal_frame.dxf` ($V_{\text{base}} = 1.757\ \mathrm{m}^3, \mathrm{DCR}_{\text{base}} = 0.846$).
- **A02 (`A02_Portal_Frame_Storage`)**: Wide-Span Portal Frame ($L = 8.0\ \mathrm{m}, H = 4.0\ \mathrm{m}$) under heavy industrial storage gravity loads ($w_{\text{trib}} = 4.5\ \mathrm{m}$) and ACI 318 checks ($V_{\text{base}} = 3.820\ \mathrm{m}^3, \mathrm{DCR}_{\text{base}} = 0.864$).
- **A03 (`A03_2Bay_2Story_Office`)**: 2-Bay 2-Story RC Moment Frame ($L = 5.0\ \mathrm{m}, H = 3.2\ \mathrm{m}$) under office gravity loads ($w_{\text{trib}} = 5.0\ \mathrm{m}$) and ACI 318 verification ($V_{\text{base}} = 5.772\ \mathrm{m}^3, \mathrm{DCR}_{\text{base}} = 0.599$).
- **A04 (`A04_2Bay_3Story_Residential`)**: 2-Bay 3-Story Frame ($L = 6.0\ \mathrm{m}, H = 3.0\ \mathrm{m}$) under residential gravity loads ($w_{\text{trib}} = 4.5\ \mathrm{m}$) and ACI 318 design ($V_{\text{base}} = 9.720\ \mathrm{m}^3, \mathrm{DCR}_{\text{base}} = 0.677$).

- **A05 (`A05_2D_Modal_Dynamics`)**: 2-Bay 3-Story Frame ($L = 6.0\ \text{m}, H = 3.5\ \text{m}$) eigenvalue modal analysis extracting fundamental periods (T_1, T_2, T_3) and mode shapes via `structdyn`.
- **A06 (`A06_2D_Response_Spectrum_ASCE7`)**: 2-Bay 3-Story Frame ($L = 6.0\ \text{m}, H = 3.5\ \text{m}$) Response Spectrum Analysis (RSA) using ASCE 7 design spectrum ($S_{DS} = 1.0g, S_{D1} = 0.6g$), CQC modal combination, gravity enveloping, and ACI 318 checks ($V_{\text{base}} = 13.309\ \text{m}^3, \text{DCR}_{\text{base}} = 0.880$).
- **A07 (`A07_2D_RSA_Ground_Motion`)**: 3-Bay 4-Story Frame ($L = 5.5\ \text{m}, H = 3.5\ \text{m}$) Response Spectrum Analysis computed from the 1994 Northridge ground motion record with CQC combination and ACI 318 checks ($V_{\text{base}} = 26.705\ \text{m}^3, \text{DCR}_{\text{base}} = 0.880$).
- **A08 (`A08_2D_Overstressed_Portal_Reflection`)**: Self-Healing Reflection benchmark on a 1-bay 1-story frame ($L = 7.0\ \text{m}, H = 3.5\ \text{m}$) deliberately initialized with undersized members ($0.25 \times 0.25\ \text{m}$ columns, $0.25 \times 0.30\ \text{m}$ beams) under heavy storage loads ($w_{\text{trib}} = 6.0\ \text{m}$), testing autonomous resizing via `modify_project` until $\text{DCR} \leq 1.0$ followed by DXF CAD generation saved to `research_paper/drawings/A08_portal_frame.dxf` ($V_{\text{base}} = 3.710\ \text{m}^3, \text{DCR}_{\text{base}} = 0.797$).

Group B: 3D Space Frames (8 Cases) — 3D Biaxial Gravity, Seismic ELF & 3D RSA

- **B01 (`B01_3D_2x2_2Story_Office`)**: 3D Space Frame (2-Story Hostel Block Plan): 2×2 bays at $6.0\ \text{m}$, 2 stories at $3.5\ \text{m}$, based on the 2-story hostel block framing plan (`research_paper/benchmarks/fixtures/hostel_block_plan_2 story.PNG`) under residential floor gravity loads, LRFD combinations, and 3D direct stiffness analysis.
- **B02 (`B02_3D_3x2_3Story_Gravity`)**: 3D Space Frame (3×2 bays at $5.0\ \text{m} \times 6.0\ \text{m}$, 3 stories at $3.2\ \text{m}$) under office gravity loads with 3D biaxial column checks ($V_{\text{base}} = 85.104\ \text{m}^3, \text{DCR}_{\text{base}} = 0.846$).
- **B03 (`B03_3D_2x2_Seismic_ELF`)**: 3D Space Frame (2×2 bays at $6.0\ \text{m}$, 3 stories at $3.5\ \text{m}$) under ASCE 7 Seismic Equivalent Lateral Force (ELF) ($S_{DS} = 1.2g, S_{D1} = 0.5g, R = 5.0$) and 3D Bresler P-M2-M3 interaction checks ($V_{\text{base}} = 143.100\ \text{m}^3, \text{DCR}_{\text{base}} = 0.983$).
- **B04 (`B04_3D_Modal_Dynamics`)**: 3D Space Frame (2×2 bays, 3 stories) 3D modal eigenvalue analysis extracting translational/torsional periods and modal mass participation factors.
- **B05 (`B05_3D_Response_Spectrum_X`)**: 3D Space Frame (2×2 bays, 3 stories) Response Spectrum Analysis along the X-direction per ASCE 7 ($S_{DS} = 1.1g, S_{D1} = 0.55g$) with CQC combination and biaxial ACI 318 checks ($V_{\text{base}} = 116.640\ \text{m}^3, \text{DCR}_{\text{base}} = 0.929$).
- **B06 (`B06_3D_Response_Spectrum_MultiDir`)**: 3D Space Frame (3×2 bays, 3 stories) multi-directional Response Spectrum Analysis (X and Y) under the 1940 El Centro spectrum with CQC combination and biaxial checks ($V_{\text{base}} = 107.700\ \text{m}^3, \text{DCR}_{\text{base}} = 0.948$).
- **B07 (`B07_3D_HighSeismic_Reflection`)**: 3D Space Frame (2×2 bays, 2 stories) Self-Healing Reflection initialized with undersized columns ($0.35 \times 0.35\ \text{m}$) under high seismic ELF loads ($\text{DCR}_0 > 1.3$), iteratively enlarging sections via `modify_project` until $\text{DCR} \leq 1.0$ ($V_{\text{base}} = 56.857\ \text{m}^3, \text{DCR}_{\text{base}} = 0.626$).
- **B08 (`B08_3D_Asymmetric_Seismic`)**: 3D Asymmetric Space Frame (3 stories) with plan irregularity and torsional eccentricity under bi-directional seismic loading, evaluating torsional drift amplification and corner column demands ($V_{\text{base}} = 113.636\ \text{m}^3, \text{DCR}_{\text{base}} = 0.940$).

Group C: Advanced Dynamic, Multimodal Vision, CAE & CAD Pipelines (8 Cases)

- **C01 (`C01_Dynamics_ElCentro_TimeHistory`)**: 2D 2-Bay 3-Story Frame dynamic earthquake time-history analysis under the 1940 El Centro record via Newmark-Beta integration in `structdyn`, peak internal force enveloping, and ACI 318 design ($V_{\text{base}} = 20.869\ \text{m}^3, \text{DCR}_{\text{base}} = 0.951$).

- **C02 (`C02_Dynamics_Northridge_TimeHistory`)**: 3D 2×2 Bay 2-Story Space Frame direct numerical time-history integration under the 1994 Northridge record, extracting peak floor acceleration, drift history, and base shear time series.
- **C03 (`C03_CAE_SAP2000_RoundTrip`)**: CSI SAP2000 CAE round-trip interoperability importing `sample_frame.s2k`, validating geometry, executing linear static FEA and ACI 318 checks, and re-exporting to `.s2k` as `research_paper/benchmarks/results/exported_sap2000.s2k` ($V_{\text{base}} = 1.710\ \text{m}^3, \text{DCR}_{\text{base}} = 0.846$).
- **C04 (`C04_CAE_ETABS_RoundTrip`)**: CSI ETABS CAE round-trip interoperability importing `sample_frame.e2k`, verifying multi-story connectivity and boundary restraints, running analysis, and re-exporting to `.e2k` as `research_paper/benchmarks/results/exported_etabs.e2k`.
- **C05 (`C05_Vision_FloorPlan_Draft_Ingestion`)**: Multimodal architectural floor plan ingestion via `struct_agent.vision` using the 3-story school plan (`research_paper/benchmarks/fixtures/floor_plan_3story_school.PNG`), compiling a 3-story 3D structural frame (`create_model_from_plan_draft`) and verifying topological connectivity.
- **C06 (`C06_Vision_To_Seismic_Design_Pipeline`)**: End-to-end multimodal floor-plan-to-seismic design pipeline on the 4-story commercial mall (`research_paper/benchmarks/fixtures/mall_plan_4story.PNG`): compiling a 3D frame from floor plan specification $\rightarrow$ ASCE 7 seismic ELF loading ($S_{DS} = 1.2g, S_{D1} = 0.5g$) $\rightarrow$ 3D FEA $\rightarrow$ ACI 318 biaxial verification ($V_{\text{base}} = 92.664\ \text{m}^3, \text{DCR}_{\text{base}} = 0.866$).
- **C07 (`C07_Seismic_TimeHistory_Dynamic_Reflection`)**: Dynamic Self-Healing Reflection on a 2-bay 3-story frame initialized with undersized 0.30×0.30 m columns under the 1940 El Centro time-history record ($\text{DCR}_0 > 1.20$), iteratively resizing columns and re-integrating equations of motion until $\text{DCR} \leq 1.0$ ($V_{\text{base}} = 20.869\ \text{m}^3, \text{DCR}_{\text{base}} = 0.951$).
- **C08 (`C08_Full_Pipeline_Autonomous_CAD_Production`)**: Full autonomous seismic design-to-CAD pipeline on a 2-bay 2-story frame ($L = 5.0$ m, $H = 3.2$ m): gravity + ASCE 7 Response Spectrum Analysis $\rightarrow$ ACI 318 design $\rightarrow$ production AutoCAD `.dxf` drawings with rebar schedules saved to `research_paper/drawings/C08_seismic_frame.dxf` ($V_{\text{base}} = 6.072\ \text{m}^3, \text{DCR}_{\text{base}} = 0.846$).

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3. Quantitative Evaluation Metrics

Each benchmark execution is evaluated across five formal metrics:

3.1 Task Completion Rate (TCR)

$$\text{TCR} = \frac{N_{\text{completed}}}{N_{\text{total}}} \times 100\% \quad (N_{\text{total}} = 24)$$

A benchmark run is counted in $N_{\text{completed}}$ if and only if the agent constructs/imports a valid `struct_core.Project` model and successfully executes all required pipeline tools specified in `expected_tools` (analysis, design, CAD drawing generation, or CAE export) without fatal runtime exceptions.

3.2 Code Compliance Rate (CCR)

$$\text{CCR} = \frac{N_{\text{compliant}}}{N_{\text{design}}} \times 100\% \quad (N_{\text{design}} = 18)$$

Evaluated across the 18 benchmark cases requiring ACI 318 structural design. A completed design is code-compliant if and only if the governing demand-capacity ratio across all structural members $e \in \mathcal{E}$ and

factored load combinations $c \in \mathcal{C}$ satisfies:

$$\mathrm{DCR}_{\mathrm{gov}} = \max_{e \in \mathcal{E},\, c \in \mathcal{C}} \left\{ \frac{P_{u,e,c}}{\phi P_{n,e}}, \frac{M_{u,e,c}}{\phi M_{n,e}}, \frac{V_{u,e,c}}{\phi V_{n,e}} \right\} \leq 1.00$$

3.3 Calculation Hallucination Rate (CHR)

$$\mathrm{CHR} = \frac{N_{\text{hallucinated_claims}}}{N_{\text{total_numerical_claims}}} \times 100\%$$

Measures the percentage of reported engineering quantities (base shear, fundamental period T_1, peak moment/shear, or governing DCR) that deviate by $> 5\%$ from the exact deterministic mechanics solution. In the proposed neuro-symbolic framework, $\mathrm{CHR} = 0.0\%$ by construction because all numerical outputs are read directly from verified symbolic tool states.

3.4 Self-Healing Convergence Iterations (N_{iter})

For cases where the initial trial design is overstressed ($\mathrm{DCR}_{\mathrm{gov}}^{(0)} > 1.0$), N_{iter} records the number of closed-loop reflection cycles (`modify_project` $\rightarrow$ `run_analysis`/`run_time_history` $\rightarrow$ `run_design`) required to achieve $\mathrm{DCR}_{\mathrm{gov}}^{(N_{\mathrm{iter}})} \leq 1.0$.

3.5 Volumetric Material Economy Ratio (η_{vol})

$$\eta_{\mathrm{vol}} = \frac{V_{\mathrm{concrete}}^{\mathrm{agent}}}{V_{\mathrm{concrete}}^{\mathrm{baseline}}}$$

Evaluated on code-compliant designs ($\mathrm{DCR}_{\mathrm{gov}} \leq 1.0$), where $V_{\mathrm{concrete}} = \sum_{e \in \mathcal{E}} A_e L_e$ is the total structural concrete volume (m^3) and $V_{\mathrm{concrete}}^{\mathrm{baseline}}$ is the verified human engineering baseline volume generated by `generate_human_baselines.py`. Values of $\eta_{\mathrm{vol}} \approx 0.95$–$1.05$ indicate that the agent achieves human-level material efficiency without gross over-sizing; when reinforcement detailing is generated (`struct_draw`), the composite cost ratio $\eta_{\mathrm{material}} = \frac{V_{\mathrm{concrete}}^{\mathrm{agent}} + \alpha W_{\mathrm{steel}}^{\mathrm{agent}}}{V_{\mathrm{concrete}}^{\mathrm{baseline}} + \alpha W_{\mathrm{steel}}^{\mathrm{baseline}}}$ is also inspected.

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4. Experimental Configurations & Multi-Model Matrix

4.1 Comparative Configurations

Configuration ID	CLI Flag (`--config`)	Description & Role in Ablation Study
Baseline 1: Direct Zero-Shot LLM	`direct_llm`	Raw LLM prompting without symbolic tools; quantifies baseline mathematical error and Calculation Hallucination Rate (CHR).
Baseline 2: Code-Generation LLM	`code_gen`	Prompts the LLM to write raw standalone Python/NumPy FEA and design scripts from scratch; measures syntax/convergence failures.
Baseline 3: Ablated Agent (No Reflection)	`ablated`	Full neuro-symbolic tool suite enabled, but closed-loop reflection (`modify_project` resizing) is strictly disabled;

Configuration ID	CLI Flag (`--config`)	Description & Role in Ablation Study
		isolates the impact of self-healing on CCR.
Proposed: Full Neuro-Symbolic Agent	`full_agent`	Complete ontology-grounded agent with multimodal perception, symbolic solvers, and autonomous self-healing reflection loop.

4.2 Evaluated LLM Backbones

Model Identifier	Architecture & Provider	Scale	Deployment Modality	Endpoint / Authentication
`gemini-3.8-flash`	Frontier Closed-Source (Google DeepMind)	Proprietary	Cloud API	`GEMINI_API_KEY` loaded from `struct_agent/.env`
`qwen3.5:9b`	Open-Weight Dense (Alibaba Qwen)	9B	Local Workstation (Ollama)	`http://localhost:11434/v1`
`gpt-oss:120b-cloud`	Open-Weight Large Model	120B	Cloud Ollama Host	`OLLAMA_HOST` environment variable
`nemotron-3-ultra:cloud`	Open-Weight Math/Reasoning-Tuned (NVIDIA)	Ultra Scale	Cloud Ollama Host	`OLLAMA_HOST` environment variable

5. Step-by-Step Reproducibility Protocol

Step 1: Activate Environment & Verify Baselines

```
conda activate structeng
python "research_paper/benchmarks/generate_human_baselines.py"
```

Step 2: Execute Full & Ablated Benchmark Campaigns

```
# 1. Google Gemini 3.8 Flash (Full Agent vs. Ablated):
python "research_paper/benchmarks/run_benchmarks.py" --model gemini-3.8-flash --
config full_agent --cases all
python "research_paper/benchmarks/run_benchmarks.py" --model gemini-3.8-flash --
config ablated --cases all

# 2. Local Qwen 3.5 9B (Full Agent vs. Ablated):
python "research_paper/benchmarks/run_benchmarks.py" --model qwen3.5:9b --config
full_agent --cases all
```

```
python "research_paper/benchmarks/run_benchmarks.py" --model qwen3.5:9b --config
ablated --cases all

# 3. Cloud GPT-OSS 120B (Full Agent vs. Ablated):
python "research_paper/benchmarks/run_benchmarks.py" --model gpt-oss:120b-cloud --
config full_agent --cases all
python "research_paper/benchmarks/run_benchmarks.py" --model gpt-oss:120b-cloud --
config ablated --cases all

# 4. Cloud Nemotron 3 Ultra (Full Agent vs. Ablated):
python "research_paper/benchmarks/run_benchmarks.py" --model nemotron-3-
ultra:cloud --config full_agent --cases all
python "research_paper/benchmarks/run_benchmarks.py" --model nemotron-3-
ultra:cloud --config ablated --cases all
```

Step 3: Compile Telemetry Tables & Publication Figures

```
python "research_paper/benchmarks/analyze_results.py"
```

This script processes all timestamped JSON logs in `research_paper/benchmarks/results/` (without overwriting prior runs) and outputs:

- `results/summary_table_<timestamp>.md`: Markdown table of TCR, CCR, N_{iter}, η_{vol}, and runtime ready for Section 5.
- `plots/fig_benchmark_comparison_<timestamp>.png`: 300 DPI grouped bar chart of TCR and CCR.
- `plots/fig_dcr_convergence_<timestamp>.png`: 300 DPI self-healing DCR vs. iteration trajectories.
- `plots/fig_material_economy_<timestamp>.png`: 300 DPI volumetric concrete economy comparison against the $\eta_{\mathrm{vol}} = 1.00$ human baseline.